STAT 641 **Homework 2** **Ghosh**

A.29 First, we introduce the inequality : $|a+b|^p \le 2^{p-1}(|a|^p + |b|^p), 1 \le p < \infty$.

a) For $1 \le p < \infty$, we have

$$\begin{aligned}(d_p(f_n, f))^p &= \int_0^1 |f_n(t) - f(t)|^p dt \\ &= \int_{1-\frac{1}{n}}^1 [1 - n(1-t)]^p dt \\ &= \frac{1}{n(p+1)} \to 0\end{aligned}$$

as $n \to \infty$. That is, $d_p(f_n, f) \to 0$ as $n \to \infty$.
But for $p = \infty$, $d_\infty(f_n, f) = 1 \nrightarrow 0$ as $n \to \infty$.

b) Since $f \in C[0,1]$, then $|f(t)| \le M$ for some $M > 0$.
For $1 \le p < \infty$, we get

$$\begin{aligned}(d_p(f_n, f))^p &= \int_0^1 |f_n(t) - f(t)|^p dt \\ &= \int_{1-\frac{1}{n}}^1 |\big(f(1-\tfrac{1}{n}) - f(t)\big)\big(1 - n(t + \tfrac{1}{n} - 1)\big) + \big(f(1) - f(t)\big)\big(n(t + \tfrac{1}{n} - 1)\big)|^p dt \\ &\le 2^{p-1} \int_{1-\frac{1}{n}}^1 |f(1-\tfrac{1}{n}) - f(t)|^p \big(1 - n(t + \tfrac{1}{n} - 1)\big)^p + |f(1) - f(t)|^p \big(n(t + \tfrac{1}{n} - 1)\big)^p dt \\ &\le 2^{p-1}(2M)^p \int_{1-\frac{1}{n}}^1 \big(1 - n(t + \tfrac{1}{n} - 1)\big)^p + \big(n(t + \tfrac{1}{n} - 1)\big)^p dt \\ &\le \frac{2^{2p} M^p}{n(p+1)} \to 0\end{aligned}$$

as $n \to \infty$. That is, $d_p(f_n, f) \to 0$ as $n \to \infty$.
For $p = \infty$, $d_\infty(f_n, f) = \sup\{|f_n(t) - f(t)| : t \in [0,1]\}$=$\sup\{|\big(f(1-\frac{1}{n}) - f(t)\big)\big(1 - n(t + \frac{1}{n} - 1)\big) + \big(f(1) - f(t)\big)\big(n(t + \frac{1}{n} - 1)\big)| : t \in [1 - \frac{1}{n}, 1]\} \le \sup\{|f(1 - \frac{1}{n}) - f(t)| : t \in [1 - \frac{1}{n}, 1]\} + \sup\{|f(1) - f(t)| : t \in [1 - \frac{1}{n}, 1]\} \to 0 + 0 = 0$ as $n \to \infty$.
Hence $d_p(f_n, f) \to 0$ for all $1 \le p \le \infty$.

A.34 First we show if O_α is open for any $\alpha \in I$, then $\cup_{\alpha \in I} O_\alpha$ is open.
For $\forall x \in \cup_{\alpha \in I} O_\alpha$, $\exists \alpha_0 \in I$ such that $x \in O_{\alpha_0}$. Since O_{α_0} is open, then $\exists \delta > 0$ such that $d(x,y) < \delta \Rightarrow y \in O_{\alpha_0} \subset \cup_{\alpha \in I} O_\alpha$, i.e. $y \in \cup_{\alpha \in I} O_\alpha$. So $\cup_{\alpha \in I} O_\alpha$ is open.

Secondly, we need to prove if A, B is open, then $A \cap B$ is open.
For $x \in A \cap B$, i.e. $x \in A$ and $x \in B$. Since A is open, then $\exists \delta_1 > 0$ such that $d(x,y) < \delta_1 \Rightarrow y \in A$. Similarly, $\exists \delta_2 > 0$ such that $d(x,y) < \delta_2 \Rightarrow y \in B$. Let $\delta = \min\{\delta_1, \delta_2\} > 0$, then $d(x,y) < \delta \Rightarrow y \in A$ and $y \in B$, i.e. $y \in A \cap B$. So $A \cap B$

is open.

Finally, we can construct an counterexample to show the intersection of an infinite number of open sets need not be open. The open set $(-\frac{1}{n}, \frac{1}{n})$ is open for any $n \geq 1, n \in N$. But the intersection $\cap_{n=1}^{\infty}(-\frac{1}{n}, \frac{1}{n}) = \{0\}$ is a closed set.

A.34 First we show if $\{f_n\}_{n\geq 1}$ converge pointwise to g on $(-1, 1)$.
For $\forall x \in (-1, 1)$, $|f_n(x) - g(x)| = |x^n| = |x|^n \to 0$ as $n \to \infty$. So $\{f_n\}_{n\geq 1}$ converge pointwise to g on $(-1, 1)$.

Secondly, we can prove that $\{f_n\}_{n\geq 1}$ converge uniformly to g on (a, b) for $-1 < a < b < 1$.
For $\forall x \in (a, b)$, then $\forall \epsilon > 0$, there exists $N_0 = N_0(\epsilon) = \max([\log_{|a|} \epsilon] + 1, [\log_{|b|} \epsilon] + 1)$ such that $\forall n > N_0$, $|f_n(x) - g(x)| = |x|^n < \max(|a|^n, |b|^n) < \epsilon$. So $\{f_n\}_{n\geq 1}$ converge uniformly to g on (a, b) for $-1 < a < b < 1$.

Finally, for $\{f_n\}_{n\geq 1}$ on (0,1), we know that $\exists \epsilon_0 > 0$, and $\forall n \in N$, there exists $x > (\epsilon_0)^{\frac{1}{n}}$ such that $|f_n(x) - g(x)| = |x|^n > \epsilon_0$. So $\{f_n\}_{n\geq 1}$ don't converge uniformly to g on $(0, 1)$

1.2 $\mathcal{F}$ is a σ-algebra, by **Definition 1.1.2**, due to the fact that any countable union of elements in $\mathcal{F}$ is essentially a finite union of elements in $\mathcal{F}$, because $\mathcal{F} \subset \mathcal{P}(\Omega)$ is finite.

1.4
1) $\forall \theta \in \Theta$, $\Omega \in \mathcal{F}_\theta \Rightarrow \Omega \in \cap_{\theta\in\Theta}\mathcal{F}_\theta$.
2) $\forall A \in \cap_{\theta\in\Theta}\mathcal{F}_\theta, \Rightarrow A \in \mathcal{F}_\theta, \forall\theta \in \Theta \Rightarrow A^c \in \mathcal{F}_\theta, \forall\theta \in \Theta \Rightarrow A^c \in \cap_{\theta\in\Theta}\mathcal{F}_\theta$.
3) $\forall \{A_n\}_{n\geq 1} \subset \cap_{\theta\in\Theta}\mathcal{F}_\theta \Rightarrow \{A_n\}_{n\geq 1} \subset \mathcal{F}_\theta, \forall\theta \in \Theta \Rightarrow \cup_{n\geq 1}A_n \in \mathcal{F}_\theta, \forall\theta \in \Theta \Rightarrow \cup_{n\geq 1}A_n \in \cap_{\theta\in\Theta}\mathcal{F}_\theta$.

1) + 2) + 3)$\Rightarrow \cap_{\theta\in\Theta}\mathcal{F}_\theta$ is a σ-algebra.

1.6 $\mathcal{F} = \{\cup_{i\in J}A_i \,:\, J \subset \mathbb{N}\}$

1) Let $J = \mathbb{N} \Rightarrow \Omega = \cup_{i\in\mathbb{N}}A_i \in \mathcal{F}$.
2) If $A \in \mathcal{F}$, then $\exists\, J \subset \mathbb{N}$ such that $A = \cup_{i\in J}A_i \Rightarrow A^c = \Omega \setminus \cup_{i\in J}A_i = \cup_{i\in J^c}A_i \in \mathcal{F}$, because $\mathcal{A} = \{A_i \,:\, i \in \mathbb{N}\}$ is a partition of Ω.
3) Suppose $\{B_n\}_{n\geq 1} \subset \mathcal{F}$, then for each $n, \exists\, J_n \subset \mathbb{N}$, such that $B_n = \cup_{i\in J_n}A_i$, then $\cup_{n\geq 1}B_n = \cup_{i\in\cup_{n\geq 1}J_n}A_i \in \mathcal{F}$, because $\cup_{n\geq 1}J_n \subset \mathbb{N}$.

1) + 2) + 3)$\Rightarrow \mathcal{F}$ is a σ-algebra.

Problem A Claim: $\sigma\langle\mathcal{C}\rangle = \mathcal{F} \equiv \{A \subseteq \Omega : A \text{ is countable or } A^c \text{ is countable}\}$.

To show $\mathcal{F}$ is a σ-algebra, note that (i) $\Omega \in \mathcal{F}$ (since $|\Omega^c| = |\emptyset| = 0$), (ii) $A \in \mathcal{F}$ implies $A^c \in \mathcal{F}$ (if A is countable, then $(A^c)^c$ is countable and if A^c is countable, then $A^c \in \mathcal{F}$ trivially). Next, suppose $\{A_i\} \subseteq \mathcal{F}$. If there exists some uncountable A_i, then A_i^c is countable, and hence $(\cup_i A_i)^c = \cap A_i^c$ is countable, so $\cup_i A_i \in \mathcal{F}$. On the other hand, if all A_i are countable, then $\cup_i A_i$ is also countable and hence belongs to $\mathcal{F}$. Thus, $\mathcal{F}$ is a σ-algebra.

On the other hand, any σ-algebra generated by $\mathcal{C}$ should contain Ω and is closed under complement and countable unions, so $\mathcal{F}$ should be the smallest among them.